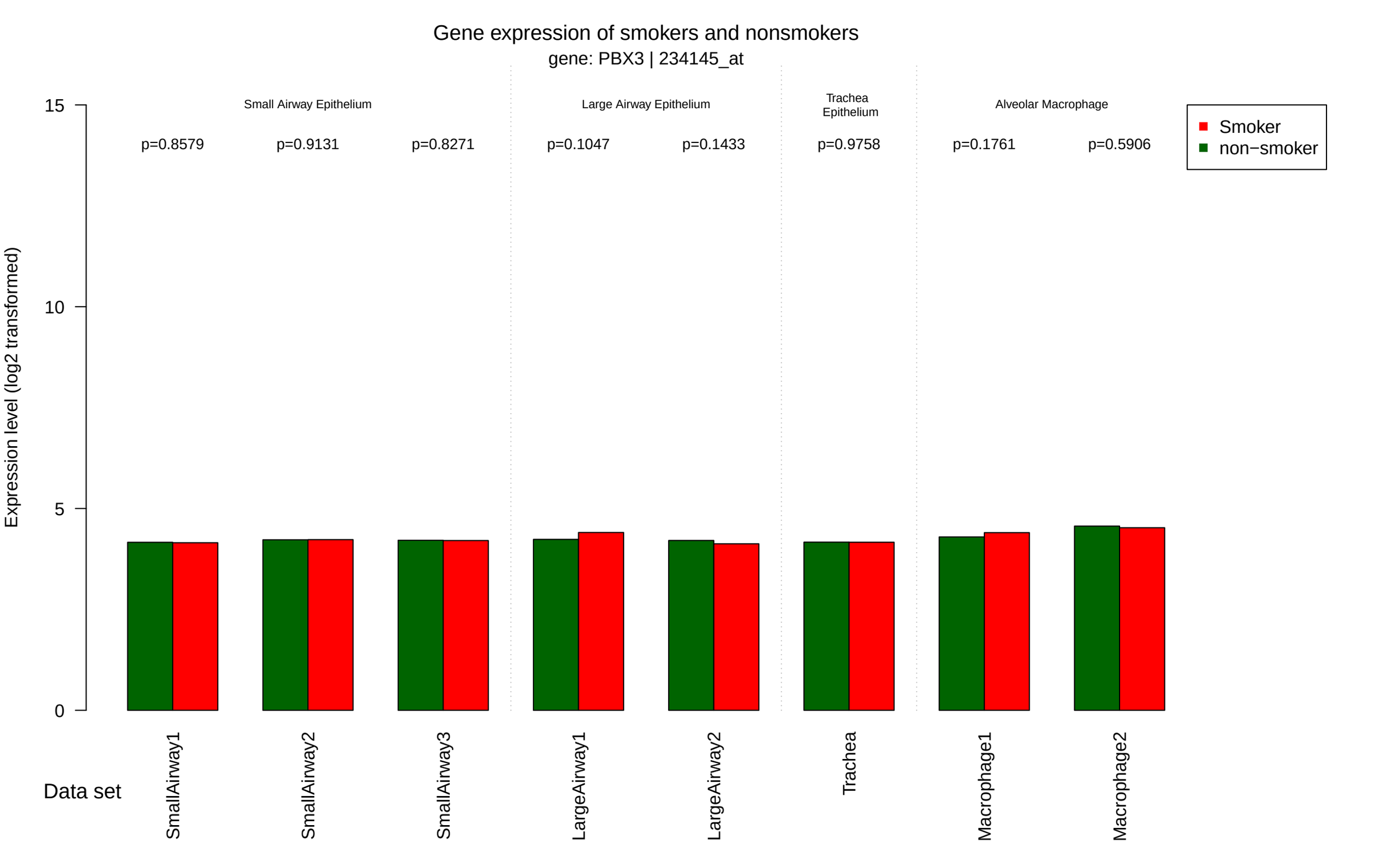
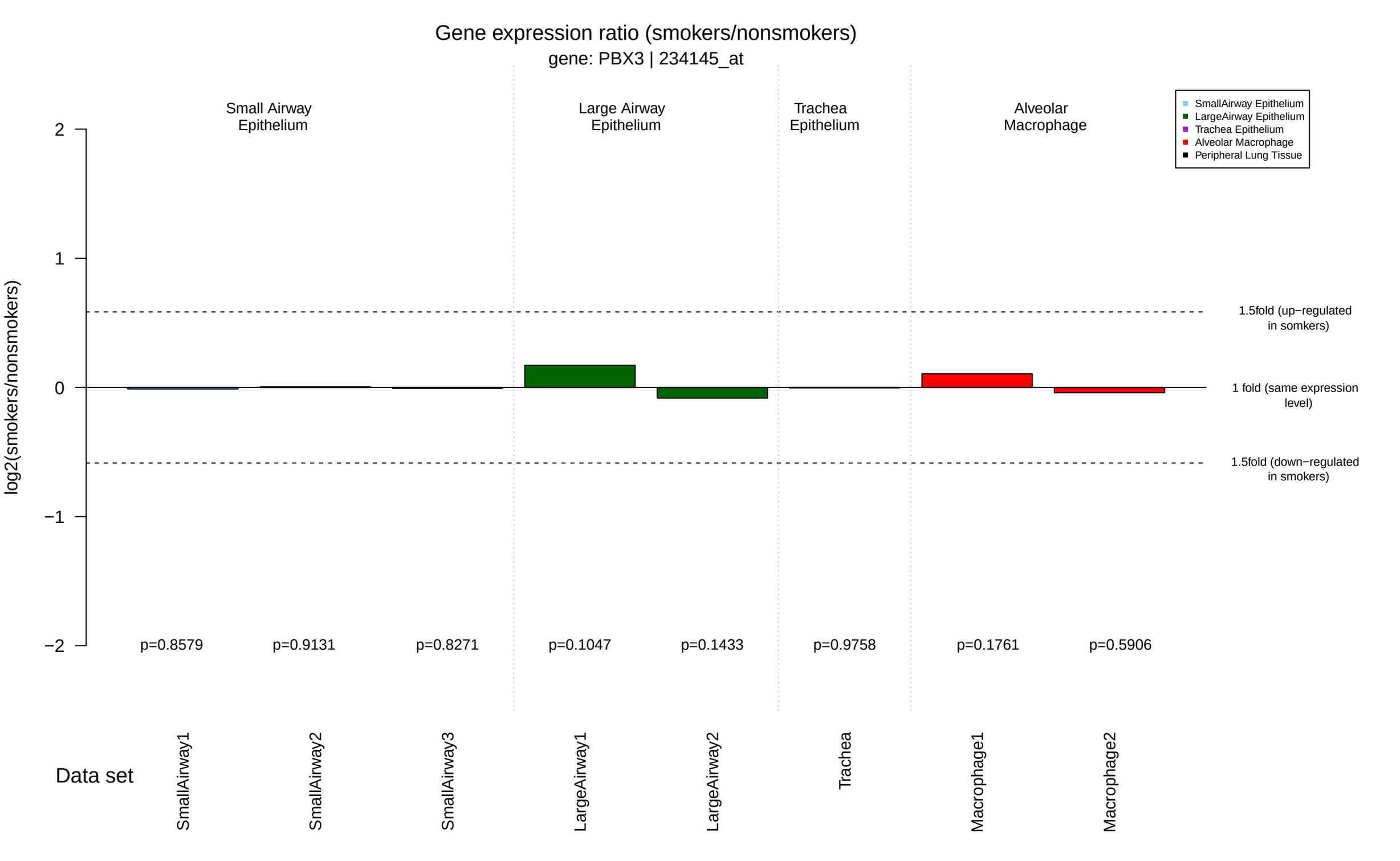
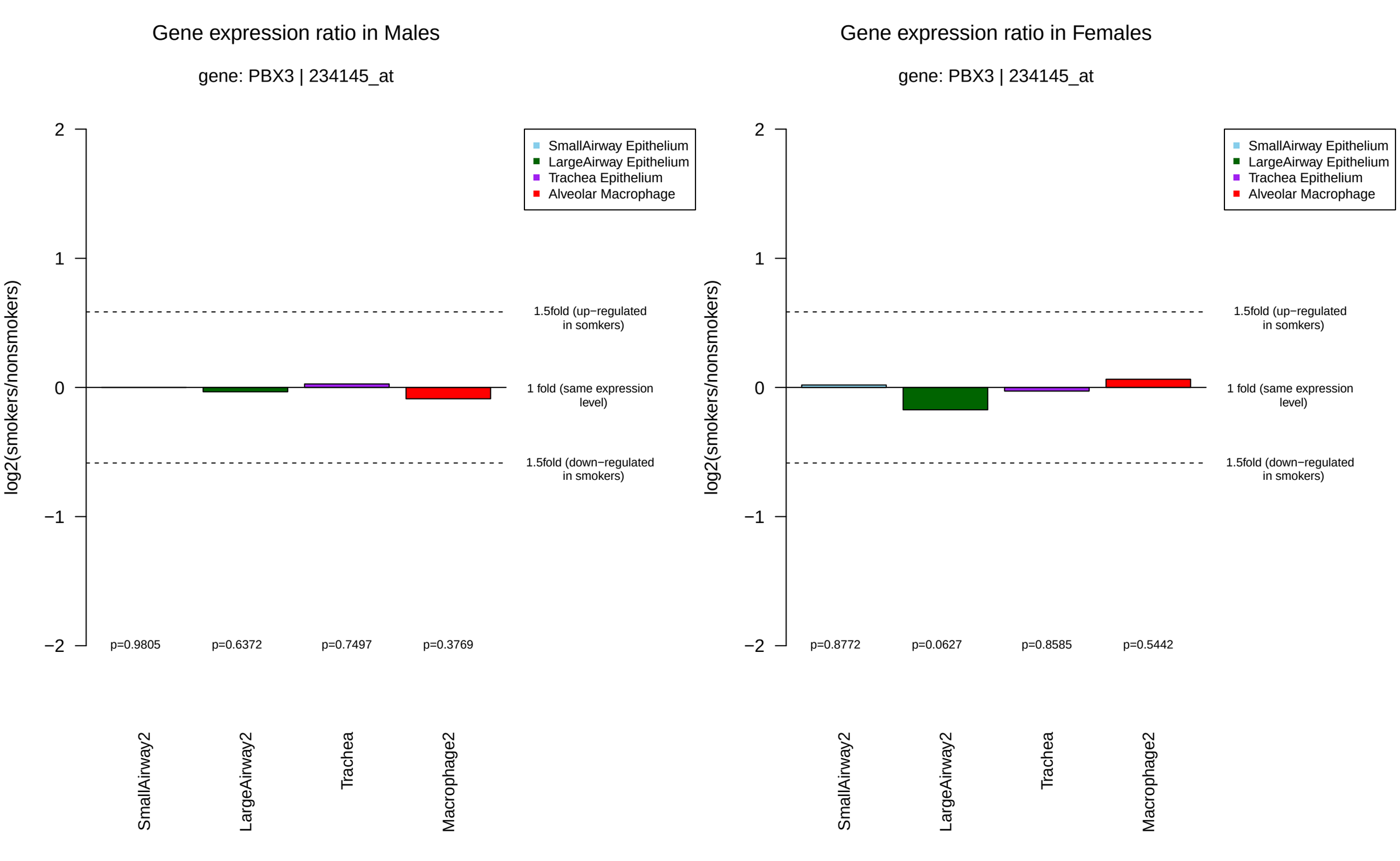
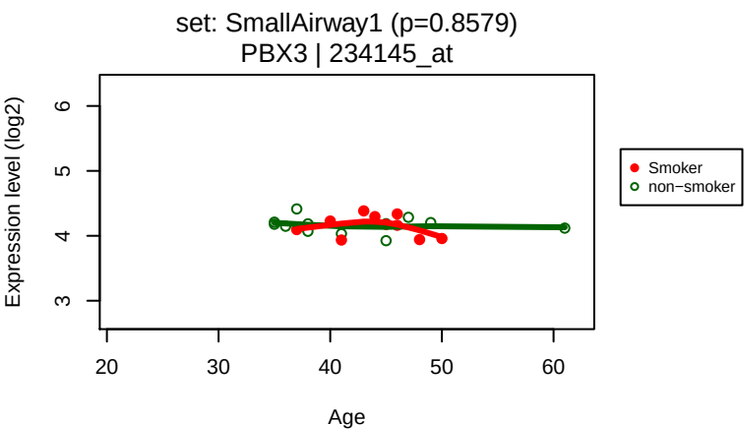
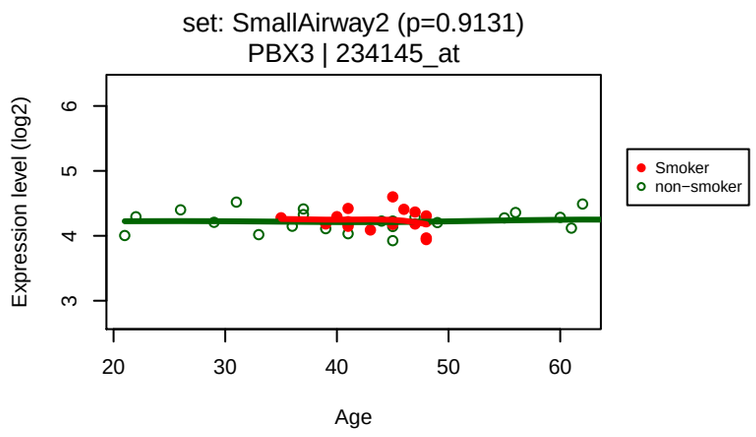
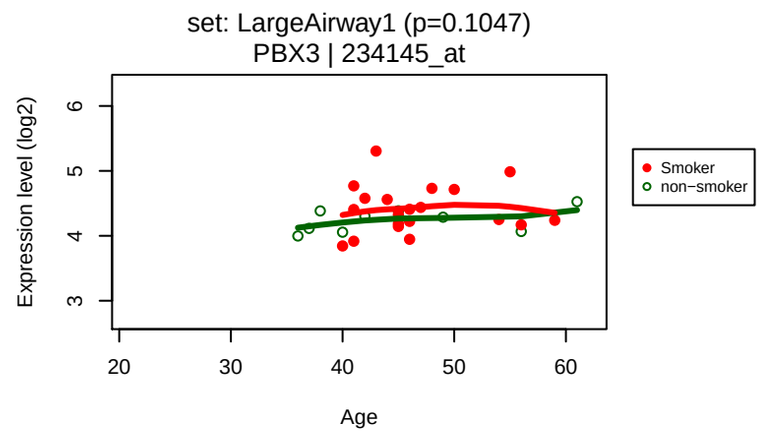

Gene expression of smokers and nonsmokers
gene: PBX3 | 234145_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8579
p=0.9131
p=0.8271
p=0.1047
p=0.1433
p=0.9758
p=0.1761
p=0.5906
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

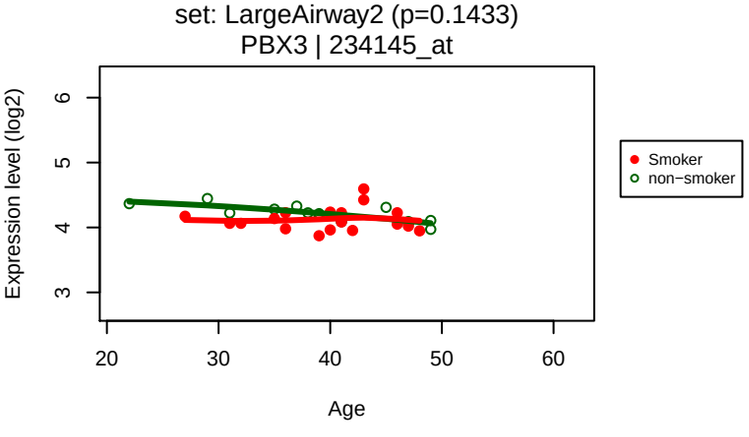

Gene expression ratio (smokers/nonsmokers)
gene: PBX3 | 234145_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8579
p=0.9131
p=0.8271
p=0.1047
p=0.1433
p=0.9758
p=0.1761
p=0.5906
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

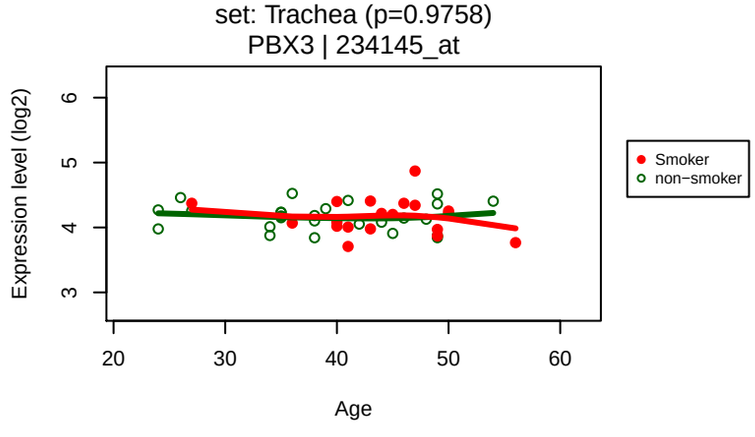

Gene expression ratio in Males
gene: PBX3 | 234145_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9805
p=0.6372
p=0.7497
p=0.3769
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PBX3 | 234145_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8772
p=0.0627
p=0.8585
p=0.5442
SmallAirway2
LargeAirway2
Trachea
Macrophage2

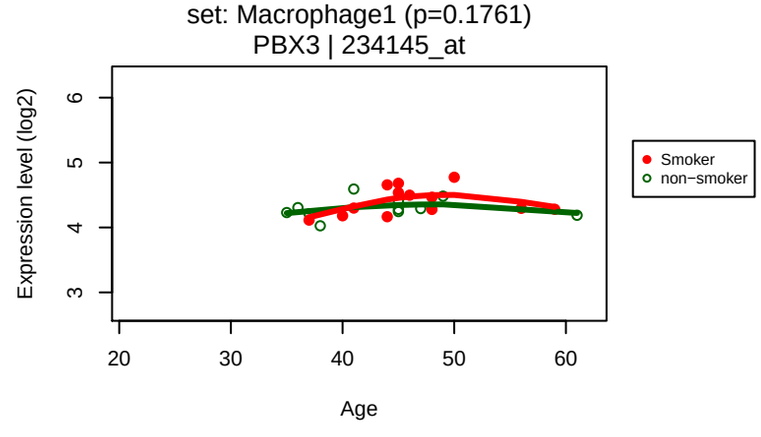

set: SmallAirway1 (p=0.8579)
PBX3 | 234145_at
Expression level (log2)
Age
Smoker
non−smoker

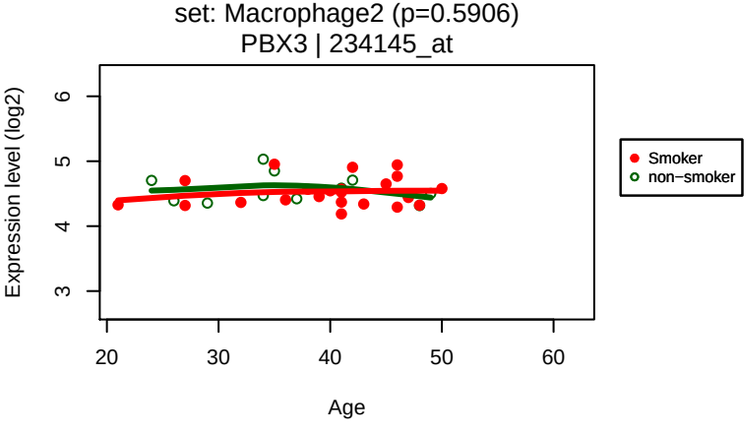

set: SmallAirway2 (p=0.9131)
PBX3 | 234145_at
Expression level (log2)
Age
Smoker
non−smoker

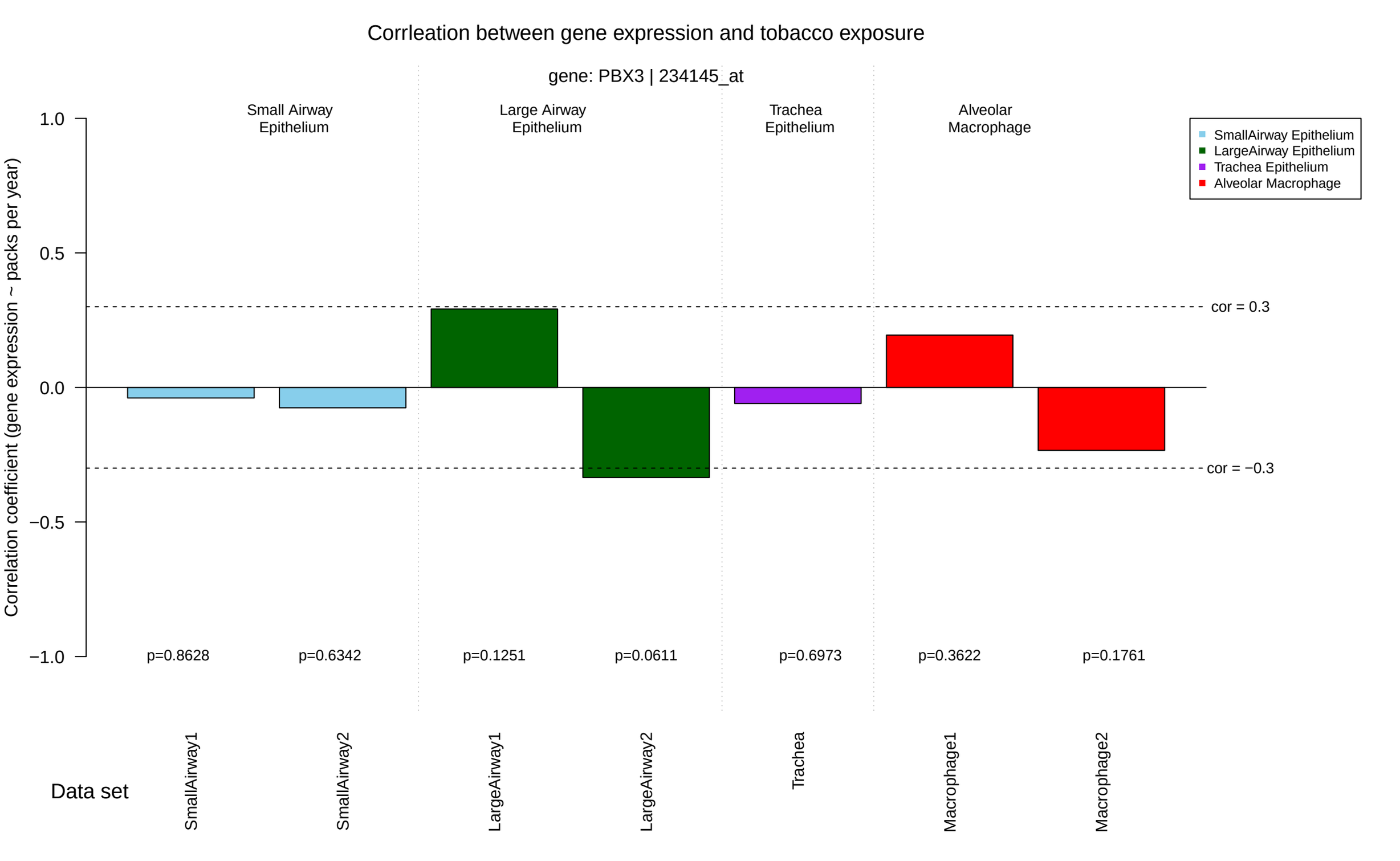

set: LargeAirway1 (p=0.1047)
PBX3 | 234145_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1433)
PBX3 | 234145_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9758)
PBX3 | 234145_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1761)
PBX3 | 234145_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5906)
PBX3 | 234145_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PBX3 | 234145_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8628
p=0.6342
p=0.1251
p=0.0611
p=0.6973
p=0.3622
p=0.1761
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2